

## MODULE 4: EMERGING DATABASE MODELS, TECHNOLOGIES AND APPLICATIONS

### Unit 4. DATABASE SERVICES AND SERVICE PROVIDERS

#### 4.4.1 Introduction

Database as a service (DBaaS) is the process of application owners paying an outside provider that launches and maintains a cloud database for storage, as opposed to having the application owners control the database themselves. Payments are per-usage and application owners can access their application data as they please. These databases provide the same functionality as a standard relational or non-relational database. DBaaS is beneficial for companies that are trying to avoid the work of configuring, maintaining, and upgrading their own databases. DBaaS lives in the overall realm of software as a service (SaaS), similar to platform as a service (PaaS) and infrastructure as a service (IaaS), where all products are hosted as a service. Other types of databases include relational database tools, NoSQL database tools, graph database tools and more. Developers on a budget also have options with free database software.

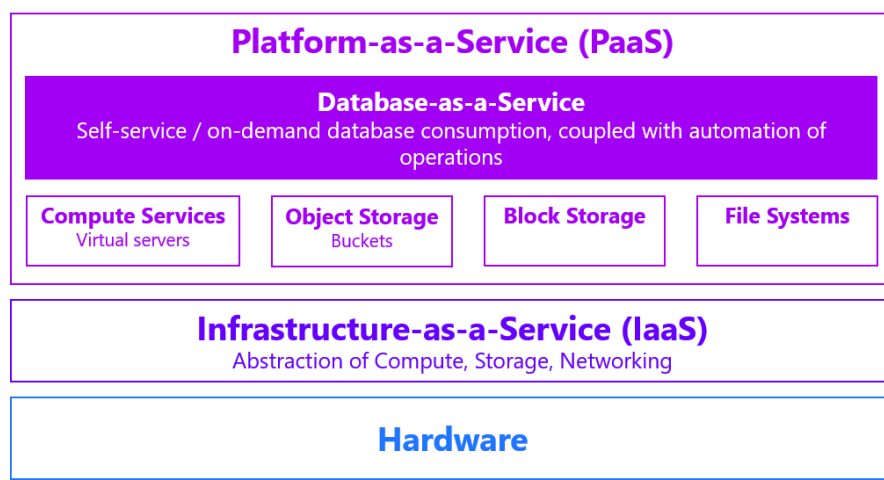
##### 4.4.1.2 Objectives:

- To understand the concept of Database as a Service
- To understand the set up and operation of DBaaS
- To understand the benefits of DBaaS

##### 4.4.1.3 Database as a Service (DBaaS)

(DBaaS) refers to software that enables users to setup, operate and scale databases using a common set of abstractions (primitives), without having to either know or care about the exact implementations of those abstractions for the specific database. The process is as follows:

1. **Setup** : Setting up a database involves provisioning a VM on which to run it, installing the database, and configuring it according to a set of parameters. Information Technology (IT) administrators managing the platform can choose to setup databases for their consumers, or enable a self-service model in which developers and DevOps create databases either through an enterprise portal, an SDK, or even using automation tools like Terraform. The self-service model has the advantage of zero IT intervention, freeing up IT admins for more important tasks. Using DBaaS, the time required to setup a database can be reduced from weeks to minutes.
2. **Operate**: Once a database has been setup, the platform is responsible for all the back-end operations to maintain it in good health. These include configuration management, automating backups (and enabling easy restore when needed), patches and upgrades, DR, service monitoring (both for the database and the underlying infrastructure) and more. All of these capabilities are provided to the IT administrator as easy single-click operations rather than the complex procedures they would have been without a DBaaS platform.
3. **Scale**: To accommodate increased usage of an application as it evolves and matures, the platform should automatically scale up database instances as needed according to a set of policies. For example, as usage grows beyond a certain threshold, data from a master instance can be automatically distributed to one or more read replica instances. Once data has been distributed over multiple instances, one of the read replicas can also be used for failover.
4. **DBaaS on IaaS**: DBaaS is often delivered as a component of a more



comprehensive platform, which may provide additional services such as Infrastructure-as-a-Service (IaaS). The DBaaS solution would request resources from the underlying IaaS, which would automatically manage the provisioning *compute, storage and networking as needed essentially removing the need for IT to be involved.*

Figure 4.4.1: DBaaS on IaaS

Source: <https://www.stratoscale.com/blog/dbaas/what-is-database-as-a-service/>

#### **4.4.1.4 Who Users DBaaS:**

It is important to understand that like other cloud technologies, DBaaS has two primary consumers:

1. The IT organization which manages and maintains the cloud
2. The end user who consumes the cloud resources, typically, developers and DevOps.

The IT organization deploys the DBaaS solution enabling end users (developers and DevOps) to provision a database of their choice, on-demand, from a catalog of supported databases, which could include both relational and non-relational databases. The IT organization can configure the DBaaS to support specific releases of these software titles, and can further restrict the configurations that specific users can provision. For example, developers may only be allowed to provision databases with a small memory footprint using traditional disks while DevOps could provision higher capacity servers with SSD's. Finally, the IT organization can setup policies for standard database operations like backups, DR and security policies to ensure that the data is properly saved from time to time to allow for recovery when required.

Typically, an end user would access the DBaaS system through a portal that offers a selection of different database titles, and in a variety of different configuration options. With a few clicks, the user specifies the required database and its corresponding

configuration for provisioning. The DBaaS system quickly provisions the database and returns a query-able endpoint such as: `mysql://192.168.15.243:3306/`. The user can then use this directly in an application. The DBaaS system provides simple mechanisms to add users, create schemas and grant permissions to different users as required by the application.

#### **4.4.2 Benefits of DBaaS**

A DBaaS solution provides an organization a number of benefits, the main ones being:

1. Developer agility
2. IT productivity
3. Application reliability and performance
4. Application security

#### **4.4.3 Developer agility**

Deploying a database is a multi-step process including the provisioning of compute, storage and networking components, configuring them properly and installing the database software. In most enterprises, this process must go through the organization's IT department and is typically, something like the following:

1. Developer opens a request in the IT ticketing system
2. Ticket sits in the queue until it gets to the top of the list according to IT priorities
3. IT evaluates ticket, and if the request is approved goes about allocating the required compute, storage and networking resources needed for the developer's database (in some cases, each of these is also handled by a separate sub-department which has its own ticketing system and set of priorities)
1. IT configures the allocated resource
2. IT installs and configures the database to utilize the underlying infrastructure according to its internal policies

3. IT provides the developer with an entry point to the database and the developer takes it from there.

Not only is this process prone to errors and omissions, it is also extremely time-consuming. The above series of actions that might be completed in hours or days, can be stretched to days, weeks and even months. This is unacceptable for developers who are trying to shorten development cycles and release faster. A DBaaS solution radically improves provisioning time by automating the process described above. The IT organization establishes the standards by which databases will be provisioned and configures the DBaaS accordingly. Once database provisioning is standardized, and the DBaaS is configured, deploying a database is a task that can be handed back to developers who can now provision databases for themselves without requiring any intervention by IT through a simple API call or a few clicks on a UI portal. This not only gives developers the agility they need by enabling them to provision databases in a “single-click self-service” model, but also, databases will always be provisioned in a consistent manner that is aligned with the best practices for that particular database.

#### **4.4.3 IT productivity**

IT is responsible for the day-two operations of the enterprise’s databases including things like tuning, configuration, monitoring, patching, upgrading, resizing periodic backups, and so on; all the things that must be done to keep databases in proper working order. As enterprises grow, and with them, the number and types of databases that must be managed and maintained, IT resources get stretched very thin (explaining the long lead time that developers have to wait before IT provisions them with a database).

DBaaS solutions enable IT to manage a much larger number of databases by treating them as “cattle” rather than “pets”. By providing abstractions for and automating tasks involved with day-two operations, a DBaaS solution vastly simplifies IT’s job, allowing operations

like upgrades and configuration changes to be done on a fleet of databases with a single action. Once relieved from micro-managing all of an enterprise's databases, IT can focus more on activities like establishing the standards of operation for the enterprise and providing quicker service for the developers who they serve.

#### **4.4.3.1 Application reliability and performance**

Modern DBaaS solutions make it easy to keep your databases highly available and running at peak performance. Through support for read replicas, in the event of a failure, the system automatically reroutes traffic to a replica ensuring system availability at all times. The system also monitors your databases to identify increased demand on resources. Using scaling policies based on resource usage thresholds, you can configure the system to automatically scale out by provisioning additional resources as demand increases, and then scale back in once demand is reduced releasing resources for other applications.

#### **4.4.3.2 Application security**

Many database engines natively provide security features such as data encryption both at rest and in transit, each using its own data structures and APIs. A DBaaS solution provides consistent management of security for all the different types of databases you might use in your organization, while adding some security features of its own. In addition to native data encryption, you might look for things like end-to-end network security with micro-segmentation, virtual private networks and security groups. A DBaaS solution might also integrate with common enterprise user stores such as LDAP and Active Directory for user authentication, and then apply fine-grained access control via different permission policies.

#### **4.4.3.3 Some DBaaS Providers:**

1. **Amazon Aurora:** MySQL and PostgreSQL-compatible relational database engine that combines the speed and availability of high-end commercial databases with the simplicity and cost-effectiveness of open source databases.
2. **IBM Db2:** Makes it easy to deploy data wherever it's needed, fluidly adapting to changing needs and integrating with multiple platforms, languages and workloads. Supported across Linux, Unix, and Windows operating systems.
3. **Amazon Relational Database Service (RDS):** A web service that makes it easy to set up, operate, and scale a relational database in the cloud. It provides cost-efficient and resizable capacity while managing time-consuming database administration tasks. Amazon RDS gives access to the capabilities of a familiar MySQL, Oracle or Microsoft SQL Server database engine. This means that the code, applications, and tools already used today with existing databases can be used with Amazon RDS. Amazon RDS automatically patches the database software and undertakes regular back ups, storing the backups for a user-defined retention period and enabling point-in-time recovery. You benefit from the flexibility of being able to scale the compute resources or storage capacity associated with your relational database instance via a single API call. In addition, Amazon RDS makes it easy to use replication to enhance availability and reliability for production databases. Amazon RDS for MySQL also enables you to scale out beyond the capacity of a single database deployment for read-heavy database workloads. As with all Amazon Web Services, there are no up-front investments required, and you pay only for the resources you use.
4. **MongoDB Atlas:** Cloud-hosted MongoDB service engineered and run by the same team that builds the database. It incorporates operational best practices we've learned from optimizing thousands of deployments across startups and the Fortune 100. Build on MongoDB Atlas with confidence, knowing you no longer need to

worry about database management, setup and configuration, software patching, monitoring, backups, or operating a reliable, distributed database cluster.

5. **Ninox Database:** Lets you integrate everything you need - applications from different departments to streamline your operations - e.g. CRM, Sales, ERP, Projects, HR & Administration.
6. **Azure SQL Database:** A relational database-as-a service using the Microsoft SQL Server Engine. SQL Database is a high-performance, reliable, and secure database you can use to build data-driven applications and websites in the programming language of your choice, without needing to manage infrastructure.
7. **Aiven:** Provides managed cloud service hosting for software infrastructure services. Acting as a central hub for all database needs, relational and non-relational database services along with a visualization suite and high-throughput message broker are offered. Service offerings includes Kafka, PostgreSQL, MySQL, Elasticsearch, Cassandra, Redis, InfluxDB and Grafana. All Aiven services are billed by the hour based on actual usage with no hidden fees.
8. **Kintone:** A no-code business application platform that allows non-technical users to create powerful apps, workflows, and databases for their teams and organizations. Using clicks instead of code, kintone users can build apps that automate business processes, collaborate on projects/tasks, and quickly report on complex data. For business users that need to get started right away, kintone also provides dozens of pre-built applications for a variety of use cases such as CRM, project management, inventory management, and much more.
9. **Google Cloud BigTable:** Google's NoSQL Big Data database service. It's the same database that powers many core Google services, including Search, Analytics, Maps, and Gmail. Bigtable is designed to handle massive workloads at consistent low latency and high throughput, so it's a great choice for both operational and analytical applications, including IoT, user analytics, and financial data analysis.



10. **IBM Cloudant:** Fully Managed — IBM Cloud service provides a fully managed, distributed JSON document database. Instantly deploy an instance, create databases, and independently scale throughput capacity and data storage to meet your application requirements. IBM expertise takes away the pain of hardware and software provisioning, patching and upgrades, while offering a 99.95 percent SLA. Secure — Cloudant is ISO27001, SOC 2 Type 2 compliant and HIPAA ready. All data is encrypted over the wire and at rest with optional user-defined key management through IBM Key Protect.
11. **Symas LMDB:** A memory-mapped DB with read performance of an in-memory database and persistence of standard disk-based databases.
12. **Vertabelo:** Vertabelo enables users to create database model, share it with team, and finally generate SQL scripts instead of writing them manually.
13. **Webair Database-as-a-Service (DBaaS):** Webair provides Database-as-a-Service (DBaaS), a reliable and secure database management solution that gives your business simple, efficient and always available access to its mission-critical data.
14. **mLab:** A fully managed cloud database service featuring automated provisioning and scaling of MongoDB databases.
15. **Alibaba Cloud ApsaraDB:** Alibaba Cloud develops highly scalable cloud computing and data management services.

### ***1. Big Data Processing and Distribution Software***

Big Data processing and distribution systems offer a way to collect, distribute, store, and manage massive, unstructured data sets in real time. These solutions provide a simple way to process and distribute data amongst parallel computing clusters in an organized fashion. Built for scale, these products are created to run on hundreds or thousands of machines simultaneously, each providing local computation and storage capabilities. Big data processing and distribution systems provide a level of simplicity to the common business

problem of data collection at a massive scale and are most often used by companies that need to organize an exorbitant amount of data. Many of these products offer a distribution that runs on top of the open-source big data clustering tool Hadoop.

#### **4.4.4.1 Big Data Processing Requirements:**

We can classify Big Data requirements based on its five main characteristics:

##### **1. Volume:**

1. Size of data to be processed is large—it needs to be broken into manageable chunks.
2. Data needs to be processed in parallel across multiple systems.
3. Data needs to be processed across several program modules simultaneously.
4. Data needs to be processed once and processed to completion due to volumes.
5. Data needs to be processed from any point of failure, since it is extremely large to restart the process from the beginning.

##### **6. Velocity:**

1. Data needs to be processed at streaming speeds during data collection.
2. Data needs to be processed for multiple acquisition points.

##### **3. Variety:**

1. Data of different formats needs to be processed.
2. Data of different types needs to be processed.
3. Data of different structures needs to be processed.
4. Data from different regions needs to be processed.

##### **1. Ambiguity:**

1. **Big Data** is ambiguous by nature due to the lack of relevant metadata and context in many cases. An example is the use of M and F in a sentence—it can mean, respectively, Monday and Friday, male and female, or mother and father.

2. **Big Data** that is within the corporation also exhibits this ambiguity to a lesser degree. For example, employment agreements have standard and custom sections and the latter is ambiguous without the right context.

1. **Complexity:**

1. Big Data complexity needs to use many algorithms to process data quickly and efficiently.
2. Several types of data need multipass processing and scalability is extremely important.
3. Processing large-scale data requires an extremely high-performance computing environment that can be managed with the greatest ease and can performance tune with linear scalability.

#### **4.4.4.2 Big Data Processing and Distribution Software:**

To qualify for inclusion in the Big Data Processing and Distribution category, a product must:

1. Collect and process big data sets in real-time
2. Distribute data across parallel computing clusters
3. Organize the data in such a manner that it can be managed by system administrators and pulled for analysis
4. Allow businesses to scale machines to the number necessary to store its data

**Some available software include:**

1. **Google BigQuery:** Google's fully managed, petabyte scale, low cost enterprise data warehouse for analytics. BigQuery is serverless. There is no infrastructure to manage and you don't need a database administrator, so you can focus on analyzing data to find meaningful insights using familiar SQL. BigQuery is a powerful Big Data analytics platform used by all types of organizations, from startups to Fortune 500 companies.

2. **Snowflake:** Cloud data platform shatters the barriers that have prevented organizations of all sizes from unleashing the true value from their data. Thousands of customers deploy Snowflake to advance their organizations beyond what was possible by deriving all the insights from all their data by all their business users. Snowflake equips organizations with a single, integrated platform that offers the only data warehouse built for the cloud; instant, secure, and governed access to their entire network of data; and a core architecture to enable many types of data workloads, including a single platform for developing modern data applications.
3. **Amazon EMR:** A web-based service that simplifies big data processing, providing a managed Hadoop framework that makes it easy, fast, and cost-effective to distribute and process vast amounts of data across dynamically scalable Amazon EC2 instances.
4. **Qubole:** Qubole is revolutionizing the way companies activate their data--the process of putting data into active use across their organizations. With Qubole's cloud-native Data Platform for analytics and machine learning, companies exponentially activate petabytes of data faster, for everyone and any use case, while continuously lowering costs. Qubole overcomes the challenges of expanding users, use cases, and variety and volume of data while constrained by limited budgets and a global shortage of big data skills. Qubole's intelligent automation and self-service supercharge productivity, while workload-aware auto-scaling and real-time spot buying drive down compute costs dramatically. Qubole offers the only platform that delivers freedom of choice, eliminating legacy lock in--use any engine, any tool, and any cloud to match your company's needs.
5. **Azure HDInsight:** HDInsight is a fully-managed cloud Hadoop offering that provides optimized open source analytic clusters for Spark, Hive, MapReduce, HBase, Storm, Kafka, and R Server.

6. **Google Cloud Dataflow:** Fully-managed service for transforming and enriching data in stream (real time) and batch (historical) modes with equal reliability and expressiveness. And with its serverless approach to resource provisioning and management, you have access to virtually limitless capacity to solve your biggest data processing challenges, while paying only for what you use.
7. **Snowplow Analytics:** Best-in-class data collection platform built for Data Teams. With Snowplow you can collect rich, high-quality event data from all your platforms and products. Your data is available in real-time and is delivered to your data warehouse of choice where it can easily be joined with other data sets and used to power BI tools, custom reports or machine learning models. The Snowplow pipeline runs in your cloud account (AWS and/or GCP), giving you complete ownership and control of your data. Snowplow frees you to ask and answer any questions relevant to your business and use case, using your preferred tools and technologies.
8. **Oracle Big Data Cloud Service:** offers an integrated portfolio of products to help organize and analyze diverse data sources alongside existing data.
9. **Cloudera:** Delivers an enterprise data cloud for any data, anywhere, from the Edge to AI. Enables people to transform vast amounts of complex data into clear and actionable insights to enhance their businesses and exceed their expectations. Cloudera is leading hospitals to better cancer cures, securing financial institutions against fraud and cyber-crime, and helping humans arrive on Mars — and beyond. Powered by the relentless innovation of the open-source community, Cloudera advances digital transformation for the world's largest enterprises.
10. **Alibaba MaxCompute:** A general purpose, fully managed, multi-tenancy data processing platform for large-scale data warehousing. MaxCompute supports various data importing solutions and distributed computing models, enabling users

to effectively query massive datasets, reduce production costs, and ensure data security.

Companies commonly have a dedicated administrator for managing big data clusters. The role requires in-depth knowledge of database administration, data extraction, and writing host system scripting languages. Administrator responsibilities often include implementation of data storage, performance upkeep, maintenance, security, and pulling the data sets. Businesses often use big data analytics tools to then prepare, manipulate, and model the data collected by these systems.

#### **4.4.5 Big Data Analytics with Oracle**

Oracle's technology provides a solution that is a fully complete analytic environment that supports full-spectrum data ingest, wrangling, data exploration & discovery through advanced and predictive analytics. It represents the combination of software, cloud computing and/or supporting hardware that has been professionally engineered, optimized, developed and deployed to support the analytic challenges faced today. A key differentiated objective is to empower analysts to explore, test, and evaluate in a self-service fashion thus reducing the need for costly programmers and data scientists.

Oracle has created a holistic, standards-based and unified approach to provide integrated analysis for all data types, analytic methods and user classes.

#### **FAST DATA**

1. With exploding intelligence data from increased number of connected devices, cyber sensors, collection platforms and social networks, there is an increase in the volumes and speed of dynamically changing data. High-velocity data brings high value, especially to national security decision-making processes. However, some of this data loses its operational value in a short time frame. Big Data allows the luxury of

time in processing for actionable insight. Fast Data, on the other hand, requires extracting the maximum value from highly dynamic and strategic intelligence. It requires processing much faster and facilitates taking timely action as close to the generated data as possible. Fast Data can also be a means of getting early answers or tip-offs of what's in your big data holdings. Industry and government needs have led to huge internal research investments at Oracle. Oracle has responded with a robust platform for Fast Data and Stream Analytics.

#### **4.4.4.1      *Oracle Stream Analytics***

The Oracle Stream Analytics platform provides a compelling combination of: an easy-to-use visual façade to rapidly create and modify Event Stream Processing applications, an advanced streaming analytical capabilities, and a comprehensive, flexible and diverse runtime platform that uses Apache Spark, Kafka and SOA streaming technology. Oracle Stream Analytics provides a pre-developed library of algorithms and detection features to leverage spatial, statistical, machine learning and well-known patterns. The abstraction and definition of streams and targets allow immediate joining of streaming data to gold copy look-up databases, and explorations that provide a stunning visual representation of real time event data.

The user has the ability to rapidly create stream exploration and analysis without writing custom code in Apache Storm. In addition to real-time event sourcing, the Oracle Stream Analytics design environment and runtime execution supports standards-based, **continuous query execution** across both event streams and persistent data stores. This enables the platform to act as the heart of intelligence for systems needing answers in microseconds to discern patterns and trends that would otherwise go unnoticed. Event processing use cases require the speed of in-memory processing with the mathematical accuracy and reliability of standard database SQL. This platform listens to incoming event

streams and executes registered queries continuously, in-memory on each event, utilizing advanced, automated algorithms for query optimization. Examples of this type of detection:

1. Correlated events: If event A happens, event B almost always follows within 2 seconds of it.
2. Missing or Out-of-Sequence events: Events A, B, C should occur in order. C is seen immediately after A, without B would be an alert.
3. Spatial/Temporal events: Target of a particular type has passed through a geo-boundary at a particular time while an attribute about the target registers positive in a watch list database.

The Oracle Stream Analytics platform also allows for both SQL and Java code to be combined to deliver robust event processing applications, leveraging standard industry terminology to describe event sources, processes, and event output or syncs. The platform provides a meta-data driven approach to defining and manipulating events within an application. Developers and analysts use a visual, directed-graph canvas and palette for application design to quickly outline the flow of events and processing across both streams and data sources. Developing the flow through drag and drop modelling and configuration wizards, the analyst can then enter the appropriate metadata definitions to connect design to implementation.

#### **4.4.6 DATA PLATFORM**

The data platform is the layer of the data management and analytic architecture where data is staged, transformed, secured and managed. The platform must provide the scalability, security and high availability that enterprises require. Oracle has focused significant R&D and innovation in this area because we realize that it is critical to all other aspects of analytics, cyber protection and mission outcomes. Oracle believes that the data platform is a unified environment, which may include an HDFS data lake, SQL and NoSQL databases,



with data transformation, data cleansing, query, and movement functions that we call the factory. All of these capabilities work in unison to bring a modern data platform to the enterprise. Where data resides, where it's staged, where it is stored, where it is managed within the platform is really dependent on numerous factors including: type, security, analysis, governance, compliance, cost and processing method. Choosing one data method, like SQL or NoSQL, for every use case is not wise. Likewise, using Hadoop/HDFS for everything is not wise. All three serve specific purposes. Oracle has invested significantly making this data platform unified. Below outlines the technologies that make up the data platform eco-system.

#### **4.4.6.1 Gold Data Using ORACLE 12c**

The gold data concept is often referred to as “gold” because of the value. Gold data is actionable. It is data that has been cleansed and is as accurate as possible. The gold data environment must have several critical features: encryption, security, high availability, survivability and transactional integrity and high performance. The platform must also be able to handle all data types such as: spatial, graph, xml, json/key value, video, imagery and relational.

All too often enterprises put in place separate databases for every data type. This causes analytical, management and security turmoil because there are separate databases for graph, spatial, relational, xml, JSON etc. Oracle 12c is a multi-model database. It is capable of providing gold data services for all of these data types in the same database. This improves analytics by creating cohesive result sets, improves security by securing all of the data together and significantly reduces the cost of all the software, infrastructure and labor compared to a multiple database gold environment.

#### **4.4.6.2 Oracle NoSQL Database**

The Oracle NoSQL Database provides network-accessible multi-terabyte distributed key/value pair storage with predictable latency. Data is stored in a very flexible key-value format, where the key consists of the combination of a major and minor key (represented as a string) and an associated value (represented as a JSON data format or opaque set of bytes). It offers full Create, Read, Update and Delete (CRUD) operations, with adjustable durability and consistency guarantees. It also provides a powerful and flexible transactional model (with ACID) that eases application development.

1. The Oracle NoSQL Database is designed to be a highly available and extreme scalable system, with predictable levels of throughput and latency, while requiring minimal administrative interaction.
2. It is also network topology and latency aware. The database driver working in conjunction with highly scalable, fault tolerant, high throughput storage engine enables a more granular distribution of resources and processing, which reduces the incidence of hot spots and provides greater performance on commodity-based hardware.

#### **4.4.6.3 Big Data SQL**

Big data's great potential has been touted for years. For many organizations, however, the challenge to making their big data vision a reality is that their ability to collect data has gotten far ahead of their ability to use it.

One of the hurdles is that an enterprise's data is typically scattered across departments, systems, and regions, and much of what is collected is of low or unknown value. Vast quantities of this data are now stored in Apache Hadoop, which makes it possible to store and process huge amounts of data at low cost. Other data is housed in relational and NoSQL databases.

The holy grail of a big data strategy is to be able to easily leverage the information in all of these data stores to get a deeper, richer, and far more valuable view of customers, business processes, and opportunities. But Hadoop programmers are in short supply, so many organizations don't have the skills they need to leverage Hadoop data. And it's not feasible to move all of that data to a single data store; the cost and the security issues are prohibitive.

Oracle Big Data SQL enables organizations to get the utmost value from data by providing queries to big data systems integrated with existing enterprise information architectures. You can then quickly leverage big data into reports or applications using existing interfaces.

In addition to simplified access and integration, Oracle Big Data SQL uses some of the proven high performance technology of Oracle Exadata and the industry-leading security features of Oracle Database to provide super-fast speed and enterprise-class security across all of your data stores.

Leveraging Oracle Exadata Smart Scan technology, Oracle's Smart Scan on Hadoop processes SQL queries at the Hadoop storage level where data is located, scans the data, and brings back only the relevant data to the end user. Less data moved means faster results—and speed is critical when leveraging big data for real-time innovations in national security processes.

On the security front, Oracle Big Data SQL extends the advanced security capabilities of the Oracle database to Hadoop and NoSQL data. With Oracle Big Data SQL, you can take advantage of proven Oracle Database security solutions for data redaction and privilege analysis, with strong controls that limit privileged user access to data.

#### **4.4.6.4 Oracle Advanced Analytics and Machine Learning**

Oracle provides several enabling technologies for data analytics, statistical analysis, time-series analysis, modelling, and machine learning requirements. These technologies can actually be used in both the product data platform and the data lab. Traditional advanced analytics are inherently weak at information technology management such as:

1. data extracts and data movement
2. data duplication resulting in no single-source of truth
3. data security exposures
4. multiple analytical tools (commercial and open source) and languages (SAS, R, SQL, Python, SPSS, etc.)

Problems become particularly egregious during a deployment phase when the worlds of data analysis and information management collide.

##### **1. In-Database Processing with Oracle Advanced Analytics**

Oracle Advanced Analytics extends the database into a comprehensive advanced analytics platform for big data analytics. With Oracle, powerful analytics are performed directly on data in the database. Results, insights, and realtime predictive models are available and managed by the database.

A data mining model is a schema object in the database, built via a PL/SQL API that prepares the data, learns the hidden patterns to build an OAA model which can then be scored via built-in OAA data mining SQL functions. When building models, Oracle Advanced Analytics leverages existing scalable technology (e.g., parallel execution, bitmap indexes, aggregation techniques) and additional developed new Oracle Advanced Analytics and Oracle Database technologies (e.g., recursion within the parallel infrastructure, IEEE float, automatic data preparation for binning, handling missing values, support for unstructured data i.e. text, etc.). The true power of embedding data mining functions within the database as SQL functions is most evident when scoring data mining

models. Once the models have been built by learning the hidden patterns in the historical data, applying the models to new data inside the database is blazingly fast. Scoring is then just a row-wise function. Hence, Oracle Advanced Analytics can “score” many millions of records in seconds and is designed to support online transactional processing (OLTP) environments.

Using Exadata’s smart scan technology it gets better. With Oracle Advanced Analytics running in an Exadata environment, SQL predicates and OAA predictive models get pushed down to the hardware layer for execution.

1. For Oracle Exadata environments, pushed to Exadata storage level for execution
2. For Oracle Big Data Appliance (Hadoop) environments, pushed to BDA storage level for execution.

The Oracle approach to data analytics is to create a platform that is open, feature rich, integrated, standards-based, and performant. It makes big data simple by integrating Hadoop, NoSQL and SQL across the Fast Data, Data Factory, Data Lake and the Data Lab environments. The ultimate goal is to provide a cost efficient, easy to use platform that transforms all data across the enterprise into actionable information that leads to better decisions. We are proud that 3rd party analysts agree with our innovations, as Gartner has chosen Oracle as the leader in Gartner’s Magic Quadrant for data platforms for data analytics.

#### ***4.4.7 Non-native Database Management Systems***

Non-native database management systems allow users of outside applications to interact with a database by retrieving data with specific query languages. Non-native database management systems provide a platform for building queries to access and administrate necessary data.

#### **4.4.7.1 Features of Non-native Database Management Systems**

To qualify for inclusion in the Non-Native Database Management System category, a product must:

1. Provide access to information stored in a database using query language
2. Offer a platform for building and executing queries

**Some features include:**

1. Connection Manager - Allow users to connect natively to the vendor's database whether on-premise or DBaaS.
2. Browser - Allow users to browse all the different database/schema objects and their properties effective management.
3. Editor - A way to create and maintain scripts and database code with debugging and integration with source control.
4. Unit Testing (Oracle) - Ensures code is functionally tested before it is released into production.
5. Static code review (Oracle) - Ensures code meets required quality level using a rules-based system.
6. SQL Optimization - Provides developers with a way to tune and optimize SQL statements and database code without relying on a DBA. Advanced optimization enables DBAs to tune SQL effectively in production.
7. Scalability testing and database workload replay - Ensures that database code and SQL will scale properly before it gets released into production.

#### **4.4.8 Some Non-native Database Management Systems**

1. **Toad For Oracle:** Toad solutions are desktop tools that give DBAs, developers and analysts a proactive, automated approach to developing and managing databases, so organizations can spend more time on strategic initiatives and less time on mundane, repetitive tasks. Quest offers several Toad editions, which support

databases such as Oracle, SQL Server, DB2, MySQL, SAP Solutions, and more, as well as offerings built specifically for database developers, DBAs and business analysts. Used by millions, Toad for Oracle is the flagship brand in the Toad portfolio, and the leading database development and optimization toolset on the market. Toad Data Point is an analyst toolset, which connects to nearly any data source and is purpose-built for an organization's data provisioning and reporting requirements. Toad solutions help organizations maximize their investments in data and database technologies by empowering database professionals to automate processes, minimize risks and cut project timelines in half.

2. **phpMyAdmin:** A free software tool written in PHP, intended to handle the administration of MySQL over the World Wide Web.
3. **Amazon Athena:** an interactive query service designed to make it easy to analyze data in Amazon S3 using standard SQL.
4. **SQLyog:** Helps developers and database administrators (DBAs) to create and manage MySQL databases with ease. SQLyog is the most complete & easy MySQL GUI tool that helps save time & increase productivity through numerous powerful features such as autocomplete, query building, query profiler, Visual Schema designer and much more.
5. **Navicat Premium:** A database administration tool that allows you to simultaneously connect to MySQL, MariaDB, SQL Server, Oracle, PostgreSQL, and SQLite databases from a single application. Navicat Premium combines functions of other Navicat products and supports most of the features used in modern database management systems, such as Stored Procedures, Events, Triggers, Functions, Views, etc.
6. **DBeaver:** A database management tool that support various databases including MySQL and PostgreSQL.

7. **DbVisualizer:** The universal database tool for developers, DBAs and analysts. It is the perfect solution since the same tool can be used on all major operating systems accessing a wide range of databases.
8. **TablePlus:** A native tool with elegant UI that allows you to simultaneously manage multiple databases such as MySQL, Postgres, SQL Server, SQLite, Microsoft SQL Server and more.
9. **Ingres:** Provides easy migration from MySQL and proprietary databases such as Oracle, SQL Server and Sybase.
10. **DBHawk:** A web-based database management and data security tool. DBHawk is compatible with various databases, including on-premises as well as databases hosted in the cloud. DBHawk provides central data security with 2FA, LDAP integration, Okta integration, data logging and auditing, and object access control. DBHawk is supported with Oracle, DB2, SQL Server, MySQL, PostgreSQL, Amazon RDS, Amazon Aurora, Azure SQL, Amazon Redshift, Greenplum, Netezza, Teradata, SAP-Hana, MariaDB, MongoDB, Hadoop, Amazon Athena (S3), and Cassandra.

These systems can be utilized by database administrators, programmers, and other employees searching for information stored in the database. Query languages often cater to specific types of databases, including DBaaS, NoSQL database tools, relational database tools and more.

#### **4.4.9 Conclusion**

Most enterprises today operate applications that require several different database technologies, a departure from recent years where the ‘corporate standard’ mandated a single database solution for all application needs. Database-as-a-Service provides a framework within which enterprises can operate all these different databases. It provides



end users with improved agility through simplified provisioning and operation, and the flexibility to choose from a number of pre-configured options established by the IT organization. DBaaS also improves the operation of fleets of diverse databases through automation and standardization allowing IT organizations to cost-effectively offer their users a number of database choices while also ensuring that these databases are operated in a safe and secure way and in compliance with established best practices.

#### **4.4.10 Tutor Marked Assignment**

1. What is big data analytics? State the requirements for big data processing
2. What are the features of Non-native Database Management Systems?
3. What a mobile database? Describe the characteristics of Mobile Database
4. List the categories three typical types of spatial queries and enumerate the features of spatial databases

#### **4.4.11 References and Further Readings**

Atzeni, P., Ceri, S., Paraboschi, S., & Torlone, R. (1999). *Database systems: concepts, languages & architectures* (Vol. 1). London: McGraw-Hill.

Batini, C., Ceri, S., & Navathe, S. B. (1992). *Conceptual database design: an Entity-relationship approach* (Vol. 116). Redwood City, CA: Benjamin/Cummings.

Connolly, T. M., & Begg, C. E. (2005). *Database systems: a practical approach to design, implementation, and management*. Pearson Education.

Elmasri, R., & Navathe, S. B. (2011). *Database systems* (Vol. 9). Boston, MA: Pearson Education.

Singh, S. K. (2011). *Database systems: Concepts, design and applications*. Pearson Education India.